Sales_Dataset

April 25, 2026

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## Sales Analysis - Business Insights

## Objective
# To analyze retail sales data, compute key business KPIs, identify trends, and␣
↪generate actionable insights using data visualization.
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## Import Libraries

# Pandas is used for data handling, while Matplotlib and Seaborn are used for␣
↪visualization.

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
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## Load Dataset

# The dataset is loaded into a Pandas DataFrame for analysis.

df = pd.read_csv("Sample_Superstore.csv", encoding='latin1')
df.head()
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   Row ID        Order ID  Order Date   Ship Date       Ship Mode Customer ID  \
0       1  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
1       2  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
2       3  CA-2016-138688   6/12/2016   6/16/2016    Second Class    DV-13045
3       4  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335
4       5  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335

     Customer Name    Segment        Country             City  …  \
0      Claire Gute   Consumer  United States        Henderson  …
1      Claire Gute   Consumer  United States        Henderson  …
2  Darrin Van Huff  Corporate  United States      Los Angeles  …
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  …
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  …

  Postal Code  Region       Product ID         Category Sub-Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases
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1       42420   South  FUR-CH-10000454        Furniture       Chairs
2       90036    West  OFF-LA-10000240  Office Supplies       Labels
3       33311   South  FUR-TA-10000577        Furniture       Tables
4       33311   South  OFF-ST-10000760  Office Supplies      Storage

                                        Product Name     Sales  Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600         2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,…  731.9400         3
2  Self-Adhesive Address Labels for Typewriters b…   14.6200         2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775         5
4                     Eldon Fold 'N Roll Cart System   22.3680         2

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
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## Inspect Dataset

# This step helps understand the structure of the dataset.

df.columns
```

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Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
       'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
       'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
       'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
      dtype='object')
```

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df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
```

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 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
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## Data Preparation

# The 'Order Date' column is converted into datetime format to enable␣
↪time-based analysis.

# Clean column names
df.columns = df.columns.str.strip()

# Convert date
df['Order Date'] = pd.to_datetime(df['Order Date'])

# Create month column
df['Month'] = df['Order Date'].dt.to_period('M')
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## Key Performance Indicators (KPIs)

# KPIs provide key business metrics such as total revenue, average order value,␣
↪and top-performing regions.

# Total Revenue
total_revenue = df['Sales'].sum()

# Average Order Value
avg_order_value = df['Sales'].mean()

# Top Regions
top_regions = df.groupby('Region')['Sales'].sum().sort_values(ascending=False)

print("Total Revenue:", total_revenue)
print("Average Order Value:", avg_order_value)
print("\nTop Regions:\n", top_regions)
```

```
Total Revenue: 2297200.8603
Average Order Value: 229.85800083049827

Top Regions:
 Region
West       725457.8245
East       678781.2400
Central    501239.8908
South      391721.9050
Name: Sales, dtype: float64
```

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## Sales Trend (Time Series)

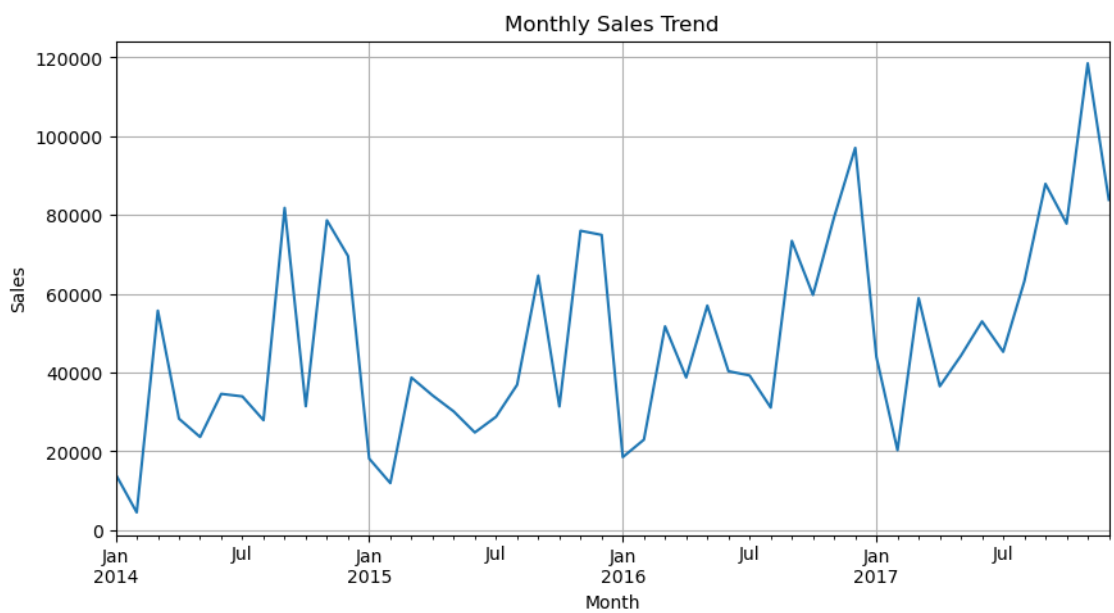
# Analyzes how sales change over time using monthly aggregation.

monthly_sales = df.groupby('Month')['Sales'].sum()

monthly_sales.plot(figsize=(10,5))

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid()

plt.savefig("sales_trend.png", dpi=300, bbox_inches='tight')
plt.show()
```



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## Top Products Analysis

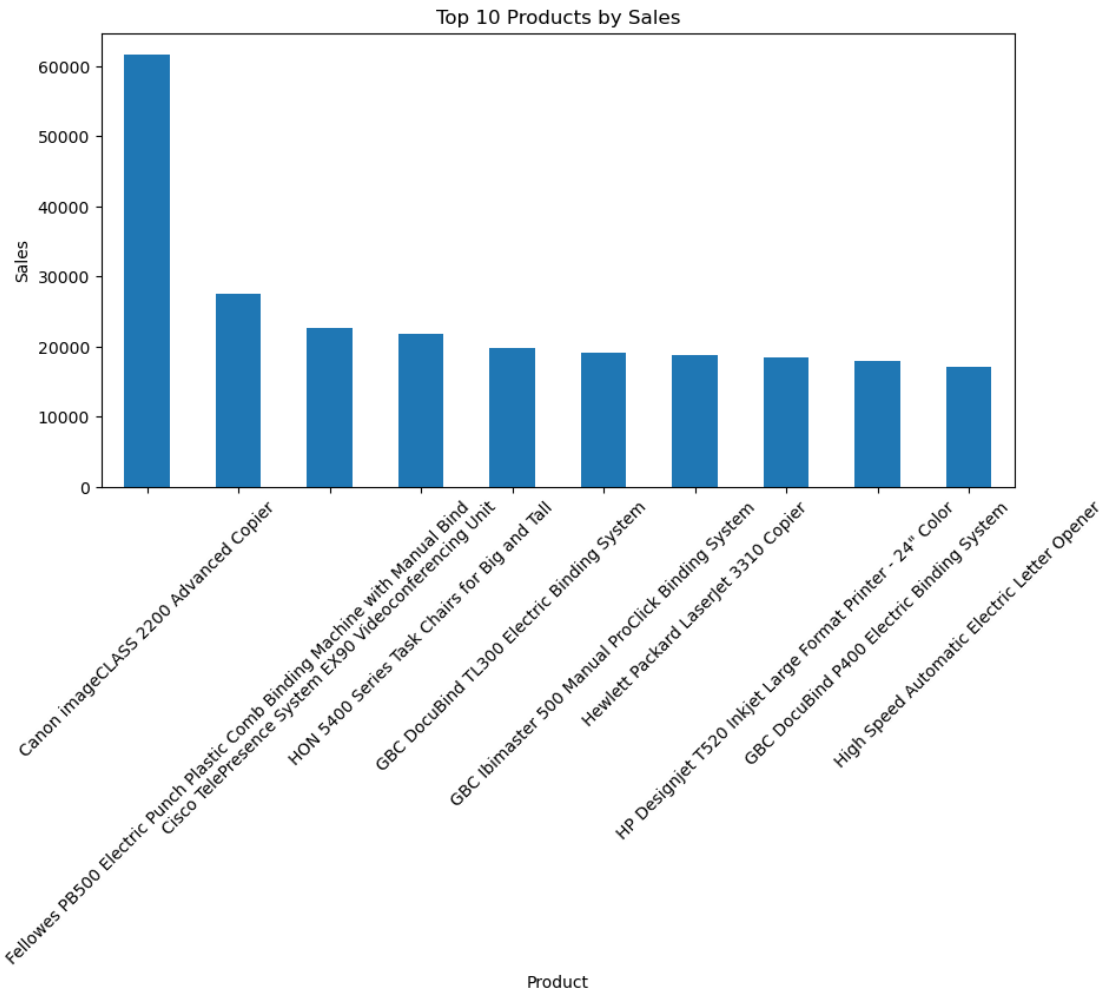
# Identifies the top-selling products based on total sales.

top_products = df.groupby('Product Name')['Sales'].sum().
 ↪sort_values(ascending=False).head(10)

top_products.plot(kind='bar', figsize=(10,5))

plt.title("Top 10 Products by Sales")
plt.xlabel("Product")
plt.ylabel("Sales")
plt.xticks(rotation=45)

plt.savefig("top_products.png", dpi=300, bbox_inches='tight')
plt.show()
```



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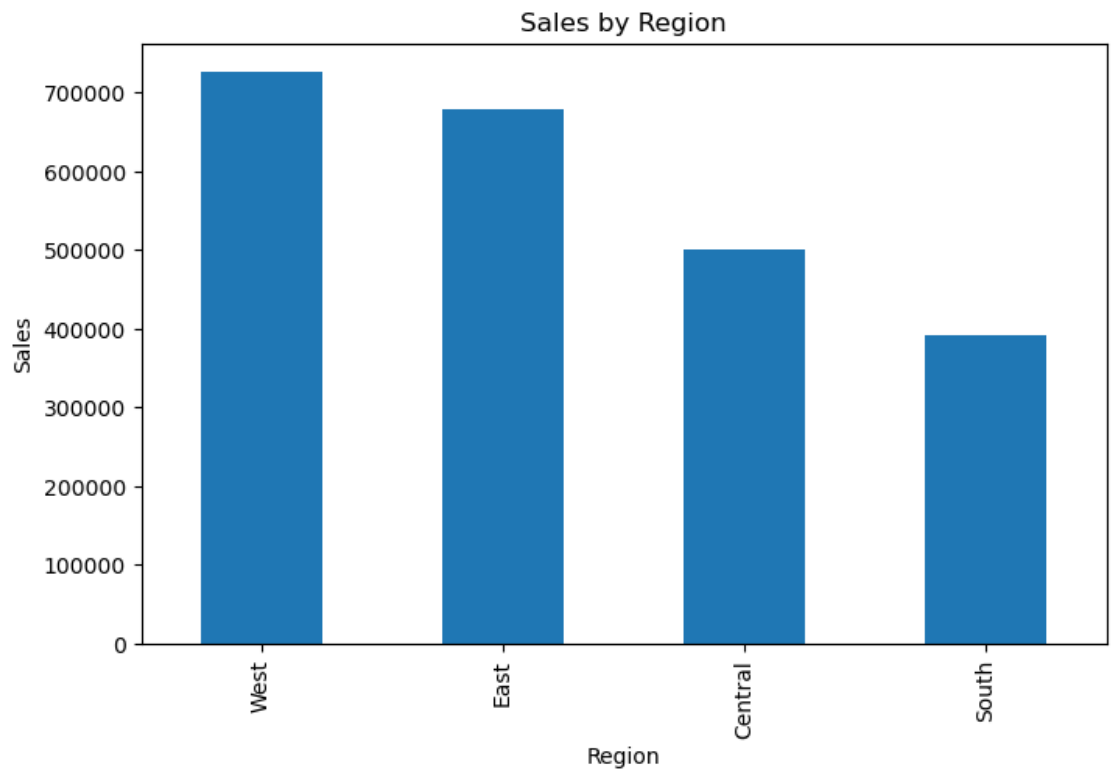
## Regional Sales Analysis

# Compares sales performance across different regions.

top_regions.plot(kind='bar', figsize=(8,5))

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

plt.savefig("region_sales.png", dpi=300, bbox_inches='tight')
plt.show()
```



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## Category Analysis

# Analyzes sales distribution across product categories.

category_sales = df.groupby('Category')['Sales'].sum()

category_sales.plot(kind='pie', autopct='%1.1f%%', figsize=(6,6))

plt.title("Sales Distribution by Category")
```

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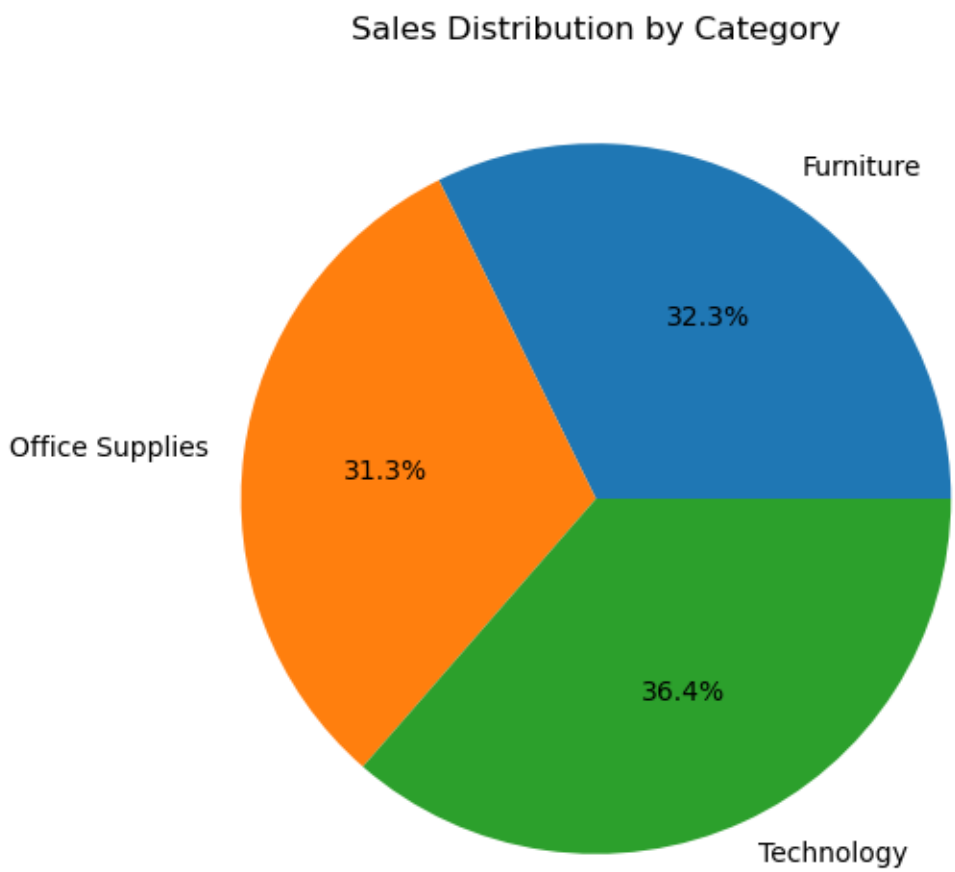
plt.ylabel("")

plt.savefig("category_distribution.png", dpi=300, bbox_inches='tight')
plt.show()
```



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## Insights

# Sales show variation across months, indicating seasonal trends
# Certain products contribute significantly to overall revenue
# Some regions outperform others in terms of total sales
# A few categories dominate the sales distribution
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## Recommendations

# Focus marketing efforts on top-performing products
# Improve sales strategies in lower-performing regions
# Plan inventory based on seasonal trends
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# Expand high-performing categories for increased revenue
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